SANKET SAHA

Senior Data Engineer · 10 Years · Databricks · Spark · Kafka · AWS
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Immediate relocation · Visa sponsorship welcome · Germany · Portugal · Netherlands

SUMMARY

Senior Software Engineer with 10 years at JP Morgan Chase, Optum, and Deloitte — specialising in large-scale Databricks Lakehouse platforms, real-time Kafka streaming, and AWS-based data infrastructure. Currently at JPMC: cut pipeline runtime by 50% (50 min → 23 min), reduced ingestion boilerplate by 90% (1,000+ lines → under 100), and built an observability platform monitoring 100+ live streaming jobs. Consistent track record of performance gains, cost savings, and platform-level engineering.

TECHNICAL SKILLS

- **Data Engineering:** Databricks, Apache Spark, PySpark, Kafka, Kinesis, Spring Batch, Delta Lake, Medallion Architecture, Streaming, ETL/ELT
- **Cloud & Infra:** AWS (ECS, EKS, S3, Lambda, Kinesis, CloudWatch, Glacier, IAM), Docker, Kubernetes, Terraform, Helm, DABS, Jenkins
- **Languages:** Python, Java 11, SQL, TypeScript, Scala
- **Backend & DB:** Spring Boot, Microservices, MySQL, Oracle, DynamoDB, MongoDB
- **Observability:** Spark UI, DAG Analysis, Splunk, Grafana, Prometheus, Spark Streaming Listener

EXPERIENCE

Senior Data Engineer (Software Engineer III) | JP Morgan Chase
Sep 2024 – Present · Hyderabad, India

- **Databricks observability** — Spark Streaming Listener captures per-job metrics (Kafka lag, data freshness, throughput) across 100+ streaming jobs processing 70+ PB/day, surfaced into live dashboards with automated alerting and ServiceNow ticket creation. Eliminated all manual monitoring; solution proposed for firm-wide adoption across JPMC.
- **Databricks Workflow Performance Improvement** — Utilising Spark UI and DAG analysis to identify bottle neck across workflows, identifying partition imbalance, task skew, and over-provisioned clusters as root causes. Fixes applied across all pipelines reduced runtime by 50% approx. (50 min to 23 min), findings formalised into a team-wide performance guide followed by every engineer onboarding new workflows.
- **Remove ingestion boilerplate by 90% (1,000+ lines → <100)** — designed a reusable Maven JAR encapsulating core ingestion logic (Kafka, API, S3, Database). New source onboarding is now fast and consistent across all applications.
- **Replacing expensive CDC problem on time-series data using Spark stateful streaming** — replaced costly Delta Lake full-dataset merges with a state-change detection layer that emits only changed records, significantly reducing compute cost and processing time.
- **Redesigned Kafka ingestion to load directly into Databricks raw tables** — removing an intermediate ECS layer, lowering infrastructure cost and pipeline complexity.
- **Multi-source enterprise Lakehouse platform** — onboarding on-premises infrastructure data from Kafka streams and REST APIs across JPMC systems into S3, structured as Delta tables (Bronze/Silver/Gold). Collaborated with data consumers, platform architects, and ops teams to define data contracts and SLAs. Cost controls via S3 Glacier tiering and DABS CI/CD.

Senior Software Lead — Data Engineering | Optum / UnitedHealth Group
Jan 2021 – Sep 2024 · Hyderabad, India

- **Processed 100,000 transactions in under 3 minutes** via Kafka + Spring Batch pipeline — processing 100,000 transactions in under 3 minutes, 4x faster than the prior system

- S**hared library strategy** — designed and published reusable Kafka and Object Storage libraries adopted by 30+ microservices, standardising data processing patterns and significantly cutting onboarding effort across the organisation.
- **AOP-based structured logging across all microservices** — The pattern to the team as a cross- cutting solution to inconsistent observability, significantly cutting incident resolution time and improving platform-wide debuggability..
- **Mentored 4 engineers** — drove adoption of clean architecture principles, raising code quality and reducing rework across the data engineering team

Consultant — Data Engineer / Full Stack | Deloitte USI
Apr 2017 – Dec 2020 · Hyderabad, India

- **Reduced SAP Order Sync failures by 90%** — resolved race conditions and data inconsistencies eliminating recurring SLA breaches.
- **Cut manual translation effort by 70% with a Python automation POC** — established a reusable pattern for multilingual processing.
- **Delivered complex features (Full Palletization, Order Management, Market Validation)** — independently across retail and financial services clients.

System Analyst | Tata Consultancy Services (TCS)
Sep 2015 – Apr 2017 · Visakhapatnam, India

- Redesigned cross-channel CCV system with server-side validation, improving security and data consistency for a major banking client.
- Led system impact analysis and enforced coding standards, reducing rework across a distributed development team.

EDUCATION

B.Tech in Information Technology · GITAM University · *2011 – 2015*

CERTIFICATIONS

- **Databricks Fundamentals Accreditation** — Databricks · Jul 2025
- **Oracle Certified Professional, Java SE 6** — Oracle · Nov 2014
- **Databricks Certified Data Engineer Associate** — In Progress
- **German Language A1** — Goethe-Institut · In Progress

LANGUAGES & PERSONAL

English — C1 · **German** — A1 in progress · **Bengali / Hindi** — Native
DOB: 16 Nov 1992 · **Nationality:** Indian · **Passport:** ZA074983 · **Location:** Hyderabad, India

ADDITIONAL

- Built a full-stack School Management System (Java + React + MySQL) — 400+ students, live attendance, fee tracking, WhatsApp parent notifications, Metabase analytics.
- Self-hosted home server running secure storage, media streaming, and home automation — hands-on DevOps beyond enterprise context.
- Hackathon participant (UHG, Deloitte). 100+ LeetCode. Hacktoberfest open-source contributor.